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6.1 INTRODUCTION

6.1.1 This chapter presents the findings of an assessment of the effects of the Proposed Development on local air quality during both the construction and post-construction stages. For both these stages, the type, source and significance of potential impacts are identified and the measures that should be employed to minimise these impacts are described.

6.1.2 A glossary of common air quality terminology is provided in **Appendix 6.1**.

6.2 ASSESSMENT METHODOLOGY

Scope

6.2.1 The scope of the assessment has been determined in the following way:

- Consultation with SBC air quality officer;
- Review of air quality data for the area surrounding the Site and background pollutant maps; and
- Review of the traffic flow data, which has been used as an input to the air quality modelling assessment.

6.2.2 The Proposed Development will give rise to the potential for impacts on local air quality during both the construction and operational stages. Details of the assessment methodology and the specific issues considered are provided below.

Data Sources

6.2.3 Air quality monitoring data used to ascertain baseline air quality in the vicinity of the Site has been obtained from the following sources:

- SBC Air Quality Review and Assessment documents, specifically the Swindon 2011 progress Report (PR2011)[1]; and
- DEFRA background maps providing estimates of background pollutant concentrations across the UK.

6.2.4 Traffic data used to predict the impact of the Proposed Development on local air quality has been provided by the transport consultant.

Assessment approach

Construction stage methodology

6.2.5 To assess the potential impacts associated with dust and PM_{10} releases during the construction stage and to determine any necessary mitigation measures, an

[1] Swindon Borough Council (2011) 2011 Air Quality Progress Report (Combined Data for 2009 & 2010) for Swindon Borough Council

assessment based on the latest guidance from the Institute of Air Quality Management (IAQM)[2] has been undertaken.

6.2.6 This approach divides construction activities into the following four categories:

- demolition;
- earthworks;
- construction; and
- trackout (the transport of dust and dirt from the construction site onto the public road network)

6.2.7 The assessment methodology then considers three separate dust effects:

- annoyance due to dust soiling;
- harm to ecological receptors; and
- the risk of health effects due to a significant increase in exposure to PM_{10}.

6.2.8 The assessment of risk of dust effects is determined by:

- the scale and nature of the works, which determine the risk of dust arising; and
- the proximity of sensitive receptors.

6.2.9 Risks are described in terms of there being a low, medium or high risk of dust effects for each of the four separate potential activities. This assessment is based on both IAQM criteria and professional judgement.

6.2.10 Mitigation measures are identified where necessary and the significance of dust effects is determined taking into account such mitigation.

6.2.11 The overall approach to the assessment of construction related dust effects is outlined in **Appendix 6.2**.

Operational stage methodology

6.2.12 The prediction of local air quality has been undertaken using the ADMS Roads dispersion model. This is a commercially available dispersion model and has been

[2] Institute of Air Quality Management, (January 2012), Guidance on the Assessment of the Impacts of Construction on Air Quality and the Determination of their Significance.

widely validated for this type of assessment and used extensively in the Air Quality Review and Assessment process.

6.2.13 The model uses detailed information regarding traffic flows on the local road network and local meteorological conditions to predict pollution concentrations at specific locations selected by the user. Meteorological data from the RAF Lyneham Meteorological Station (the closest suitable meteorological site to the Proposed Development) for 2012 has been used for the assessment.

6.2.14 The model has been used to predict road specific concentrations of oxides of nitrogen (NO_x) and particulate matter (PM_{10}) at selected receptors. The predicted concentrations of NO_x have been converted to NO_2 using the LAQM calculator on the DEFRA air quality website[3].

6.2.15 A summary of the traffic data used in the assessment can be found in **Appendix 6.3**. The data includes details of annual average daily traffic flows (AADT), vehicle speeds and percentage HGV for the assessment years considered.

6.2.16 The most recent emission factors released by DEFRA in June 2012, provided in the emissions factor toolkit EFT2012_5.1.2 and built into the ADMS model (Version 3.1), have been used to predict existing traffic emissions.

6.2.17 To predict local air quality, traffic emissions predicted by the model must be added to local background concentrations. Background concentrations of NO_2, PM_{10} and $PM_{2.5}$ have been taken from the recently released 2010 DEFRA background maps. The maps provide an estimate of background concentrations between 2010 and 2030. Data for 2011 has been used to predict existing and future pollution concentrations at the Site and adjacent to the surrounding road network. The background concentrations used in the assessment are provided in Table 6.8 in section 6.4.

6.2.18 Emission factors and background data used in the prediction of future air quality concentrations predict a gradual decline in pollution levels over time due to improved emissions from new vehicles and the gradual renewal of the vehicle fleet. However, recent monitoring carried out in urban areas throughout the UK has found that NO_2 concentrations are not declining as rapidly as previously anticipated. Monitoring

[3] http://uk-air.defra.gov.uk

carried out in Swindon has shown considerable variation in concentrations in recent years with no pattern of overall decline or increase emerging. Therefore, as a cautious approach and to ensure a worst case assessment of future air quality, concentrations of NO_2, PM_{10} and $PM_{2.5}$ in 2026 (the assumed year of completion of the Proposed Development) have been assumed to be the same as the measured 2011 background concentrations.

6.2.19 It is recommended, following guidance set out in LAQM.TG(09), that the model results are compared with measured data to determine whether the model results need adjusting to more accurately reflect local air quality. This process is known as verification.

6.2.20 LAQM.TG(09) recommends that model predictions should be within 25% (preferably 10%) of monitored concentrations for the model to be predicting with any degree of accuracy. There are two monitoring sites located adjacent to the road network being considered within this assessment. Both are diffusion tube sites. Site S6 is a roadside site located adjacent to Queens Drive in the centre of Swindon. Site S22 is a background site located adjacent to the A419.

6.2.21 LAQM.TG(09) suggests that background monitoring sites should only be used for verification where the model has been used to predict background concentrations. However, due to the close proximity of the S22 site to the A419, it is considered likely that concentrations recorded at this site are influenced by emissions from the road, it is therefore not considered to represent true background concentrations. Additionally, some of the selected receptors used in the modelling are located a similar distance from the A419. It is therefore considered appropriate to use this monitoring site for verification and adjustment of the model results.

6.2.22 The results of the comparison between modelled and monitored concentrations of NO_2 are presented below in Table 6.1.

Table 6.1: Comparison of Modelled and Monitored Nitrogen Dioxide Concentrations (µg/m^3)

Monitoring Locations	Measured Concentration	Modelled Concentration	% Difference
S6	**37.2**	**18.9**	**-49.3**
S22	**21.4**	**16.3**	**-24.0**

6.2.23 The comparison of monitored and modelled concentrations indicates that the model is under-predicting annual mean NO_2 concentrations by an average of 37% compared to monitored concentrations. It is therefore considered necessary to adjust the model results to better represent local concentrations. The results of the modelling assessment have been adjusted using the methodology given in LAQM.TG(09). Full details of the verification and calculation of adjustment factors are provided in **Appendix 6.4**.

6.2.24 There is no suitable monitoring of PM_{10} or $PM_{2.5}$ data to allow verification of the PM model results. However, LAQM.TG(09) suggests applying the NO_x adjustment factor to modelled road-PM_{10} where no appropriate verification against PM_{10} data can be carried out. Therefore, the adjustment applied to predicted NO_x concentrations has also been applied to the modelled PM_{10} and $PM_{2.5}$ concentrations.

6.2.25 LAQM.TG(09) does not provide a method for the conversion of annual mean NO_2 concentrations to 1-hour mean NO_2 concentrations. However, research[4] has concluded that exceedances of the 1-hour mean objective are generally unlikely to occur where annual mean concentrations do not exceed 60 µg/m^3. Care has been taken to ensure that locations where the 1-hour mean objective is relevant are included in the assessment.

6.2.26 Quantitative assessment of the impacts on local air quality from road traffic emissions associated with the operation of the Proposed Development have been completed against the current statutory standards and objectives set out in **Appendix 6.5** for NO_2 and PM_{10}.

Sensitive Receptors

6.2.27 LAQM.TG(09) describes in detail typical locations where consideration should be given to pollutants defined in the Regulations. Generally, the guidance suggests that all locations "where members of the public are regularly present" should be considered. At such locations, members of the public will be exposed to pollution over the time that they are present, and the most suitable averaging period of the pollutant needs to be used for assessment purposes.

[4] D Laxen and B Marner: Analysis of the relationship between 1-hour and annual mean nitrogen dioxide at UK roadside and kerbside monitoring sites (July 2003).

6.2.28 For instance, on a footpath, where exposure will be transient (for the duration of passage along that path), comparison with short-term standards (i.e. 15 minute mean or 1 hour mean) may be relevant. In a school or adjacent to a private dwelling, however, where exposure may be for longer periods, comparison with long-term standards (such as the 24 hour mean or annual mean) may be most appropriate. In general terms, concentrations associated with long-term standards are lower than short-term standards owing to the chronic health effects associated with exposure to low level pollution for longer periods of time.

6.2.29 To assess the suitability of the Site for development with regards to air quality, pollution concentrations have been predicted at four receptors located at the boundary of the Site, representing worst case exposure to air quality. The locations of these receptors, numbered A to D, are shown in **Appendix 6.6**.

6.2.30 The impact of the Proposed Development on local air quality has been assessed by predicting the change in pollution concentrations at 31 locations adjacent to the road network considered in the assessment. The selected receptors represent worst case exposure to air quality at residential and education facilities within the area. Details, including the OS Grid reference for each receptor, are provided below and their locations shown in **Appendix 6.6**.

6.2.31 Details of these receptors are:

1. Nightingale Farm – 416570, 181515;
2. Meadow Way – 419350, 180968;
3. Old Cottages – 419927, 180858;
4. 43 Purley Road – 420111, 181772;
5. Matley Moor – 419205, 182527;
6. Nightbridge House – 418875, 182566;
7. Moorleaze House – 419441, 182702;
8. 59 White Edge Moor – 419343, 182904;
9. Partridge Close – 419538, 184696;
10. 61 Church Ground – 419196, 185461;
11. Nythe Farm – 419055, 185663;
12. Canal Cottage – 419622, 186886;
13. Longleaze Farm House – 420396, 187129;

14. Manor Cottages – 419771, 187588;
15. 1 Church Ground – 419571, 187810;
16. Cambel Cottage – 419318, 187960;
17. The Wickets – 419289, 188004;
18. Yew Tree Gardens – 419276, 187806;
19. Heard & Son Fencing – 418971, 186883;
20. 292 Ermin St – 417880, 186834;
21. Coleraine – 418280, 186367;
22. Keble Close – 418023, 185454;
23. 3 Falconscroft – 418301, 185555;
24. 118 Sydney Close – 416855, 185271;
25. Logan Close – 416278, 184802;
26. 69 Graham Street – 415713, 184907;
27. The Frying Fish (residential at second floor) – 415962, 184866;
28. 317 Queens Drive – 416824, 183814;
29. 527 Queens Drive – 417066, 183314;
30. St Joseph's College – 416501, 185067;
31. St Joseph's School (Diffusion tube S6) – 416075, 184913

Significance Criteria

Construction Stage

6.2.32 The significance of effects predicted during the construction stage of the Proposed Development is based on criteria set out in the IAQM guidance. In the first instance, the sensitivity of the area and receptors being assessed are established based on the criteria set out in Table 6.3.

6.2.33 As detailed in the assessment methodology, each activity during construction is assessed and the risk of the activity giving rise to dust effects is determined as high, medium or low. The risk of dust effects can then be assessed against the sensitivity of the area/receptors, as defined in Table 6.3, to ascertain the significance of effects for each activity. However, as the implementation of best practice mitigation is standard practice following the IAQM guidance, it is recommended that the significance of effects is assessed post-mitigation, using the criteria set out in Table 6.2 below.

6.2.34 The significance of residual impacts is addressed at paragraph 6.6.4 below, being based on the initial assessment which identifies the potential effect (as summarised in Table 6.9) and assuming the mitigation measures set out in section 6.6 will be implemented through the proposed Construction Environmental Management Plan (CEMP), which would be secured through a condition on any grant of Outline planning permission for the Proposed Development.

Table 6.2: Significance of Effects for Each Activity with Mitigation

Magnitude of Effect	**Sensitivity of area/receptors**			
	Very High	**High**	**Medium**	**Low**
High	Minor Adverse	Minor Adverse	Negligible	Negligible
Medium	Minor Adverse	Negligible	Negligible	Negligible
Low	Negligible	Negligible	Negligible	Negligible
Negligible	Negligible	Negligible	Negligible	Negligible

Table 6.3: Factors Defining Sensitivity of Area and Receptors

Sensitivity of area/receptors	Receptor Type	
	Human Receptors	**Ecological Receptors[1]**
Very High	Very densely populated area, more than 100 dwellings within 20 m Local PM_{10} concentrations exceed the objective Contaminated buildings present Very sensitive receptors (e.g. oncology units) Works continuing in one area of site for more than a year	European designated site
High	Densely populated area 10-100 dwellings within 20m of site Local PM_{10} concentrations close to the objective (e.g. annual mean 36-40 $\mu g/m^3$) Commercially sensitive horticultural land within 20m	Nationally Designated site
Medium	Suburban or edge of town area Less than 10 receptors within 20m Local PM_{10} concentrations below the objective (e.g. annual mean 30-36 $\mu g/m^3$)	Locally Designated site
Low	Rural area or industrial area No receptors within 20m Local PM_{10} concentrations well below the objectives (less than 75%) Wooded area between site and receptors	No designations

[1] only if they are habitats that might be sensitive to dust

Post-completion Stage

6.2.35 Guidance issued by Environmental Protection UK (EPUK) relating to Air Quality considerations within the planning process was updated in April 2010[5]. In this guidance, criteria are detailed which identify the need for an Air Quality Assessment, the type of Air Quality assessment required and the significance of any predicted impact.

6.2.36 The above guidance sets out criteria for assessing air quality impact magnitude. This updated guidance also places greater emphasis on judging overall impacts by means of professional judgement, whilst taking into account the impact magnitude descriptors.

6.2.37 The determination of impact significance within this assessment has considered both the impact descriptors and the professional judgement of the author. Tables 6.4 and 6.5 identify the assessment criteria from this guidance.

6.2.38 These criteria have been deemed suitable for use in this appraisal, as there are currently no standards or statutory significance criteria available for this purpose.

Table 6.4: Definition of Impact Magnitude for Changes in ambient Annual Mean NO_2 and PM_{10}

Magnitude of Change	Annual Mean NO_2 and PM_{10}	Annual Mean $PM_{2.5}$
Large	Increase/ decrease >4µg/m^3	Increase/ decrease >2.5µg/m^3
Medium	Increase/ decrease 2 – 4 µg/m^3	Increase/ decrease 1.25 – 2.5 µg/m^3
Small	Increase/ decrease 0.4 – 2 µg/m^3	Increase/ decrease 0.25 – 1.25 µg/m^3
Imperceptible	Increase/ decrease <0.4 µg/m^3	Increase/ decrease <0.25 µg/m^3

[5] EPUK (April 2010), Development Control: Planning for Air Quality (2010 Update).

6.2.39 Once the magnitude of the impact is known, the next step is to describe the impact. The EPUK guidance therefore presents a set of descriptors as a means of describing impacts. These are identified in Table 6.5.

Table 6.5: Air Quality Impact Descriptors for Changes to Annual Mean NO_2 and PM_{10} Concentrations at a Receptor

Absolute Concentration in Relation to Objective/ Limit Value	Change in Concentration		
	Small	**Medium**	**Large**
Increase with Development			
Above Objective/ Limit Value *With* Scheme (>40 µg/m^3)	Minor Adverse	Moderate Adverse	Major Adverse
Just Below Objective/ Limit Value *With* Scheme (36-40 µg/m^3)	Minor Adverse	Moderate Adverse	Moderate Adverse
Below Objective/ Limit Value *With* Scheme (30-36 µg/m^3)	Negligible	Minor Adverse	Minor Adverse
Well Below Objective/ Limit Value *With* Scheme (<30 µg/m^3)	Negligible	Negligible	Minor Adverse
Decrease with Development			
Above Objective/ Limit Value *Without* Scheme (>40 µg/m^3)	Minor Beneficial	Moderate Beneficial	Major Beneficial
Just Below Objective/ Limit Value *Without* Scheme (36-40 µg/m^3)	Minor Beneficial	Moderate Beneficial	Moderate Beneficial
Below Objective/ Limit Value *Without* Scheme (30-36 µg/m^3)	Negligible	Minor Beneficial	Minor Beneficial
Well Below Objective/ Limit Value *Without* Scheme (<30 µg/m^3)	Negligible	Negligible	Minor Beneficial

6.2.40 Other factors taken into account in determining the significance of the impacts predicted are summarised in Table 6.6 below.

Table 6.6: Factors Taken into Account in Determining Air Quality Significance

Factors
The number of properties affected by minor, moderate or major air quality impacts.
The number of people exposed to levels above the objective or limit value.
The magnitude of the changes and the description of the impacts at relevant receptors.
Whether or not an exceedence of an objective or limit value is predicted to arise in the Study Area where none existed before, or an exceedence area is substantially increased.
Whether or not the study area exceeds an objective or limit value and this exceedence is removed or the exceedence area is reduced.
Uncertainty, including the extent to which worst-case assumptions have been made.
The extent to which an objective or limit value is exceeded, e.g. an annual mean NO_2 of 41 µg/m^3 should attract less significance than an annual mean of 51 µg/m^3.

Uncertainties and limitations

6.2.41 The assessment has relied on data provided by SBC and DEFRA to characterise baseline conditions in the vicinity of the Site. It has been assumed that this data has been reported correctly.

6.2.42 The modelling has not accounted for any variations in topography and therefore has assumed the Site and surrounding road network is flat.

6.2.43 The assessment of traffic emissions has been carried out using the ADMS Roads model. No computer based model will be able to totally replicate actual conditions and therefore some inaccuracies may have been introduced into the assessment.

6.2.44 The prediction of future air quality relies on emission factors that have been developed based on research, vehicle emission testing and assumed changes in the national vehicle fleet in future years. This may introduce some inaccuracies and limitations to the prediction of future air quality.

6.2.45 Operational traffic impacts have been assessed using predicted development traffic data supplied by the Applicant's traffic consultants. Any uncertainties in/limitations to the traffic data will therefore have followed through to the modelling assessment.

6.3 RELEVANT POLICY

Air Quality Strategy for England, Scotland, Wales and Northern Ireland

6.3.1 The Government's policy on air quality within the UK is set out in the Air Quality Strategy (AQS) for England, Scotland, Wales and Northern Ireland (AQS) published in July 2007[6], pursuant to the requirements of Part IV of the Environment Act 1995. The AQS sets out a framework for reducing hazards to health from air pollution and ensuring that international commitments are met in the UK. The AQS is designed to be an evolving process that is monitored and regularly reviewed.

6.3.2 The AQS sets standards and objectives for ten main air pollutants to protect health, vegetation and ecosystems. These are benzene (C_6H_6), 1,3-butadiene (C_4H_6), carbon monoxide (CO), lead (Pb), nitrogen dioxide (NO_2), particulate matter (PM_{10}, $PM_{2.5}$), sulphur dioxide (SO_2), ozone (O_3) and polycyclic aromatic hydrocarbons (PAHs).

6.3.3 The air quality standards are long-term benchmarks for ambient pollutant concentrations which represent negligible or zero risk to health, based on medical and scientific evidence reviewed by the Expert Panel on Air Quality Standards (EPAQS) and the World Health Organisation (WHO). These are general concentration limits, above which sensitive members of the public (e.g. children, the elderly and the unwell) might experience adverse health effects.

6.3.4 The air quality objectives are medium-term policy based targets set by the Government which take into account economic efficiency, practicability, technical feasibility and timescale. Some objectives are equal to the EPAQS recommended standards or WHO guideline limits, whereas others involve a margin of tolerance, i.e. a limited number of permitted exceedances of the standard over a given period.

6.3.5 For some pollutants there is both a long-term (annual mean) standard and a short-term standard. In the case of NO_2, the short-term standard is for a 1-hour averaging period, whereas for PM_{10} it is for a 24-hour averaging period. These periods reflect the varying impacts on health of differing exposures to pollutants (e.g. temporary exposure on the pavement adjacent to a busy road, compared with the long term exposure of residential properties adjacent to a road).

[6] The Air Quality Strategy for England, Scotland, Wales and Northern Ireland – July 2007.

6.3.6 The current statutory standards and objectives are set out in the table presented in **Appendix 6.5**.

6.3.7 Of the pollutants included in the AQS, NO_2, PM_{10} and $PM_{2.5}$ will be particularly relevant to this Proposed Development as these are the primary pollutants associated with road traffic.

Local Air Quality Management (LAQM)

6.3.8 Part IV of the Environment Act 1995 also requires local authorities to periodically review and assess air quality within their administrative areas. The reviews have to consider the present and future air quality and whether any air quality objectives prescribed in Regulations are being achieved or are likely to be achieved in the future.

6.3.9 Where any of the prescribed air quality objectives are not likely to be achieved the authority concerned must designate that part an Air Quality Management Area (AQMA).

6.3.10 For each AQMA, the local authority has a duty to draw up an Air Quality Action Plan (AQAP) setting out the measures the authority intends to introduce to deliver improvements in local air quality in pursuit of the air quality objectives. Local authorities are not statutorily obliged to meet the objectives, but they must show that they are working towards them.

6.3.11 The Department for Environment, Food and Rural Affairs (DEFRA) has published technical guidance for use by local authorities in their Review and Assessment work[7]. This guidance, referred to in this chapter as LAQM.TG(09), has been used where appropriate in the assessment.

Control of dust and particulates associated with construction

6.3.12 Section 79 of the Environmental Protection Act (1990) states that where a statutory nuisance is shown to exist, the local authority must serve an abatement notice. Statutory nuisance is defined as:

[7] Department for Environment, Food and Rural Affairs (DEFRA), (2009): Part IV The Environment Act 1995 Local Air Quality Management Review and Assessment Technical Guidance LAQM.TG(09).

- "any dust or other effluvia arising on industrial, trade or business premises and being prejudicial to health or a nuisance"; and
- "any accumulation or deposit which is prejudicial to health or a nuisance".

6.3.13 Failure to comply with an abatement notice is an offence and, if necessary, the local authority may abate the nuisance and recover expenses.

6.3.14 In the context of the Proposed Development, the main potential for nuisance of this nature will arise during the construction stage - potential sources being the clearance, earthworks, construction and landscaping processes.

6.3.15 There are no statutory limit values for dust deposition above which 'nuisance' is deemed to exist - 'nuisance' is a subjective concept and its perception is highly dependent upon the existing conditions and the change which has occurred. However, research has been undertaken by a number of parties to determine community responses to such impacts and correlate these to dust deposition rates.

National Planning Policy Framework (March 2012)

6.3.16 The NPPF identifies 12 core planning principles that should underpin both plan-making and decision-taking, including a requirement for planning to "contribute to conserving and enhancing the natural environment and reducing pollution" (para. 17)

6.3.17 Under section 11: Conserving and Enhancing the Natural Environment, the Framework requires the planning system to "prevent both new and existing developments from contributing to or being put at unacceptable risk or being adversely affected by unacceptable levels of air pollution" (para. 120).

6.3.18 In dealing specifically with air quality, the Framework states that "planning policies should sustain compliance with and contribute towards EU limit values or national objectives for pollutants, taking into account the presence of Air Quality Management Areas and the cumulative impacts on air quality from individual sites in local areas. Planning decisions should ensure that any new development in Air Quality Management Areas is consistent with the local air quality action plan" (para. 124).

Swindon Borough Local Plan 2011 (2006)

6.3.19 There are no 'saved' policies within the Swindon Borough Local Plan that are relevant to the control of emission.

Draft Swindon Borough Local Plan 2026

6.3.20 Under Policy EN5, all developments must be assessed for their potential impact on environmental amenity including tranquillity, noise, pollution and light pollution.

6.3.21 Policy EN7 deals specifically with pollution and states that all development that is likely to lead to emissions of pollutants, including pollutants emitted to air, will only be permitted where emissions are controlled to ensure no significant loss of amenity occurs for existing land uses or habitats.

6.4 BASELINE CONDITIONS

Swindon Borough Council Review and Assessment of Air Quality

6.4.1 As a result of the 2011 Progress Report (PR), SBC identified a requirement to carry out a Detailed Assessment (DA) of air quality due to exceedances of the annual mean NO_2 objective of 40 µg/m^3, although no AQMAs have been declared to date. The exceedances of the objective were found along Kingshill Road and Devizes Road, which are located in the south of the town, approximately 5 km south west of the Site. Air quality in the environs of the Site has not been found to be exceeding the air quality objectives.

Automatic Local Monitoring Data

6.4.2 SBC does not carry out any monitoring within the Borough using automatic monitors.

Non-automatic Monitoring

6.4.3 NO_2 is measured at 23 locations within the Borough using diffusion tubes. However, there are no such monitoring locations in the environs of the Site. The closest locations are approximately 3 km to the west and south of the Site, adjacent to Queens Way (A4312) and the A419, respectively. Details of both sites and data recorded between 2008 and 2011 are presented below in Table 6.7.

6.4.4 All the diffusion tube data has been bias adjusted using a factor derived from the latest version of the national database of co-location studies (version 03/12) available on the LAQM Support website[8]. A factor of 0.85 was applied to the 2010 monitoring data, a factor of 0.79 to the 2009 data and 0.87 to the 2008 data.

6.4.5 The diffusion tube data identified in Table 6.7 shows that annual mean NO_2 concentrations are below the annual mean objective at roadside and background locations. At the roadside site, concentrations show considerable annual variability with high concentrations recorded during 2008 and 2011. Concentrations declined significantly between 2008 and 2009, falling by over 9 µg/m^3, although concentrations increased again in 2010, with a further increase in 2011.

6.4.6 The background site adjacent to the A419 also recorded the highest concentrations during 2008, falling by over 5 µg/m^3 in 2009 to 19 µg/m^3. Concentrations remained

[8] http://laqm.defra.gov.uk/bias-adjustment-factors/national-bias.html

similar during 2010, with only a slight increase compared to 2009. The data indicates background concentrations in the area 'well below' the annual mean objective.

Table 6.7: Diffusion Tube Monitoring Results (µg/m^3)

Site	Classification	OS grid Ref	Distance from Site (approximate)	Year			
				2008	2009	2010	2011
S6 – Chalet School, Queens Drive	Roadside	416075, 184913	3 km west	39.5	30.3	33.6	37.2
S22 – 39 Edale Moor, Liden	Back-ground	419355, 183078	3 km south	25.9	19.3	21.4	-

6.4.7 There is no monitoring of PM_{10} or $PM_{2.5}$ carried out in the Borough.

DEFRA Background Maps

6.4.8 Additional information on background concentrations in the environs of the Site has been obtained from the DEFRA background pollutant maps. The average background concentrations from the following grid squares, which represent the Site, are provided in Table 6.8:

- 419500, 187500;
- 420500, 187500;
- 421500, 187500; and
- 420500, 188500.

6.4.9 New 2010 background maps have recently been published by DEFRA. These maps provide estimated background concentrations between 2010 and 2030. Concentrations for 2011, the base year and 2026, the assumed year of completion of the Proposed Development, are provided.

Table 6.8: Estimated background concentrations from DEFRA maps (µg/m^3)

Pollutant	2011	2026
NOx	24.3	15.7
NO_2	13.6	10.9
PM_{10}	16.6	14.9
$PM_{2.5}$	11.1	9.6

6.4.10 The data presented in Table 6.8 shows that estimated annual mean background concentrations of NO_2, PM_{10} and $PM_{2.5}$ are 'well below' the annual mean objectives for each pollutant.

The projected future baseline

6.4.11 Data presented in Table 6.8 indicates a significant decline in background pollution concentrations between 2011 and 2026. However, current monitoring data recorded throughout the UK has found NO_2 and PM_{10} concentrations are not declining as rapidly as previously expected. By 2026, it is anticipated that pollutant concentrations will have declined compared to existing levels, but not to the extent estimated by the DEFRA background maps.

6.5 POTENTIAL EFFECTS

Construction stage

6.5.1 The Site covers an area of approximately 1.6 km^2; therefore the Proposed Development is considered to be 'large', following the IAQM guidance. There are also sensitive receptors in close proximity to the Site. It is therefore considered necessary to assess the risk of dust effects occurring.

6.5.2 The assessment of the Site's risk category is based primarily on the following factors:

- the scale and nature of the works;
- the proximity of receptors; and
- professional judgement.

6.5.3 The large majority of the Site is currently used for agricultural purposes. There are residential properties located to the north of the Site within the village of South Marston. The closest residential properties to the Site are those located to the north of Nightingale Lane and to the south of Thornhill Road on Ash Gardens, Manor Park, Rawlings Close and Bell Gardens. There are also various residential properties located along Old Vicarage Lane, which adjoin the Site, as well as a number of properties to the west along Thornhill Road. A significant number of these properties lie within 20-40 m of the Site boundary.

6.5.4 The closest ecologically sensitive sites to the Site are Tuckmill Meadows Site of Special Scientific Interest (SSSI), approximately 2.6 km to the north-east and Stanton Park Local Nature Reserve, approximately 2.2 km to the north-west. Dust emissions are unlikely to result in significant impacts on ecologically sensitive sites beyond 100 m from the emission source as detailed in the IAQM guidance. The potential for adverse impacts from construction activities to occur at the adjacent sensitive ecological receptors is therefore considered to be negligible and has not been considered any further within this assessment.

6.5.5 The precise behaviour of the dust, its residence time in the atmosphere, and the distance it may travel before being deposited will depend upon a number of factors. These include wind direction and strength, local topography and the presence of

intervening structures (buildings etc.) that may intercept dust before it reaches sensitive locations. Furthermore, dust would be naturally suppressed by rainfall.

6.5.6 A windrose from RAF Lyneham (2012) is provided in **Appendix 6.7**, which shows that the prevailing wind is from the south-west. Properties to the north and east of the Site are therefore more likely to experience impacts than those to the south and west.

Demolition

6.5.7 There are a number of buildings located within the site boundary the majority of which will be retained. However, it is understood that a number of agricultural buildings associated with Rowborough Farm will be demolished. The nearest residential receptor to Rowborough Farm is Brookland, over 150 m to the southwest, although Rowborough Farm House lies within 20 m of the farm buildings that will be demolished.

6.5.8 The buildings to be demolished have a building volume of < 20,000 m^2 therefore the site is considered to have an emissions class of ‘small’. Due to the close proximity of Rowborough Farm House to the agricultural buildings there is considered to be a ‘medium’ risk of impacts as a result of demolition, however at all other properties the risk of impacts will be ‘low/negligible’...

Earthworks

6.5.9 Earthworks will primarily involve excavating material, haulage, tipping and stockpiling. This may also involve levelling of the Site and landscaping.

6.5.10 The Site is approximately 1.6 km^2 in size; therefore, it would be classed as being large with regard to earthworks. However, the Proposed Development will be built in phases, with approximately 200 residential units built per year. Earthwork activities would therefore only occur over small parts of the Site at any given time giving rise to less than 10 heavy earth moving vehicles on site at any given time. However, excavated material is likely to be stored on site prior to removal, which may include the formation of bunds of more than 8m in height. The Site is therefore classed as being 'medium/large' based on the IAQM guidance.

6.5.11 Due to the proximity of sensitive receptors, i.e. within 20-40 m of the Site boundary and the presence of residential properties that will be retained in the immediate environs of the Site, the Site is considered to be a 'medium/high' risk site during earthwork activities, prior to mitigation.

Construction

6.5.12 Dust emissions during construction will depend on the scale of the works, method of construction, construction materials and duration of build.

6.5.13 The Proposed Development seeks Outline permission; therefore detailed information is unavailable regarding the method and duration of construction. However, as discussed previously, approximately 200 residential properties could be constructed per year, resulting in a construction period of around 12 years. The Proposed Development will also include retail, leisure and educational facilities. The total volume of buildings constructed on the Site will therefore be > 100,000 m^3. It is anticipated that a considerable proportion of the building materials used on site will be concrete, a potentially dusty material, and that piling and on-site concrete batching will also be required. The Site is therefore classed as being 'large' and given the close proximity of receptors to the Site boundaries and that a number of residential properties will be retained in the immediate environs of the Site, there is considered to be a ‘high’ risk of impacts occurring during the construction stage, prior to mitigation.

Trackout

6.5.14 Factors influencing the degree of trackout and the associated magnitude of effect include vehicle size, vehicle speed, vehicle numbers, geology and duration.

6.5.15 As a general guidance, significant trackout may occur up to 500 m from large sites, 200 m from medium sites and 50 m from small sites, as measured from the site exit.

6.5.16 It is anticipated that approximately 75 dwellings could be actively constructed on the Site at any one time. It is therefore anticipated that there will be in the region of 10 HGV vehicles accessing the site per day, i.e. 20 HGV trips per day. The magnitude of effect is therefore considered to be 'small', based on the IAQM guidance.

6.5.17 The route to be taken by construction vehicles accessing the Site is not known at this stage; however, access to the Site for all vehicles will be along the A420 and Old Vicarage Lane. There are only a few residential properties located adjacent to both roads; therefore, as significant trackout is only anticipated to occur within 50 m of the Site entrance, the Site is considered to have a 'low' risk of impacts as a result of trackout based on the IAQM guidance.

Summary of Dust/PM_{10} Risk Effects

6.5.18 A summary of the potential dust risk effects prior to mitigation are presented in Table 6.9 below.

Table 6.9: Summary of Dust/PM_{10} Risk Effects Prior to Mitigation

Source	Dust Soiling Effects	PM_{10} Effects
Demolition	**Medium/Low Risk**	**Negligible**
Earthworks	**Medium/High Risk**	**Medium/Low Risk**
Construction	**High Risk**	**Medium Risk**
Trackout	**Low Risk**	**Negligible Risk**

Construction Traffic and Plant Emissions

6.5.19 During construction of the Proposed Development, lorries would require access to the Site to deliver and remove materials; earthmoving plant and other mobile machinery would work on-site and generators and cranes would be in operation. These machines would produce exhaust emissions such as fine particulate matter. The majority of construction vehicles would be diesel powered, which emit significantly more PM_{10} than petrol vehicles. Due to the nature of the operations, the vehicles may also be operating at the extremes of their capacity, when emissions would be greater than at optimum working conditions. When added to the small proportion of PM_{10} originating from the construction activities, these sources can result in locally elevated PM_{10} concentrations.

6.5.20 Construction of the Proposed Development would be carried out in phases with approximately 75 dwellings being built at any one time. This would require up to 75 tradesmen on site which would generate up to 150 two way vehicle trips per day. In addition, it is anticipated that there will be 20 two-way HGV movements per day during the construction period. It is therefore anticipated that there will be a maximum of 170 additional two-way vehicle movements per day on the adjacent road network during the construction period.

6.5.21 The EPUK air quality guidance sets out criteria to assist in establishing when an air quality assessment will be required. These criteria indicate that significant impacts on air quality are likely to occur where a development results in greater than 200 HGV movements per day during a construction period of a year or more. Therefore it has been concluded that construction traffic generated by the Proposed Development would result in a negligible impact on local NO_2, PM_{10} and $PM_{2.5}$ concentrations.

Post-completion stage

NO_2 Concentrations

6.5.22 Annual mean NO_2 concentrations predicted at the Site are presented below in Table 6.10, while concentrations predicted at existing receptors are provided in Table 6.11.

6.5.23 The modelling assessment shows that predicted annual mean NO_2 concentrations are 'well below' the annual mean objective of 40 µgm^3 across the Site. The impact of the Proposed Development with regard to exposure of future occupants to poor air quality is therefore considered to be negligible.

6.5.24 Annual mean NO_2 concentrations predicted at the existing receptors are also 'well below' the annual mean objective in both the base year and with Proposed Development scenarios.

6.5.25 The maximum increase in NO_2 concentrations as a result of the Proposed Development is predicted to be 2.4 µg/m^3. (at receptor 12 Canal Cottage, a residential property located adjacent to the A420). This is considered to be a medium change in air quality following the EPUK guidance (Table 6.4). However, as annual mean NO_2 concentrations are predicted to be 'well below' the objective limit, a medium change in air quality is considered to be of negligible significance.

6.5.26 At all the remaining receptors, the impact of the Proposed Development is predicted to be <1 µg/m^3, a small/imperceptible change in concentrations, with a negligible significance.

6.5.27 Exceedence of the 1-hour objective for NO_2 is also unlikely based on the predicted annual mean concentrations. Guidance referred to earlier in this chapter indicates that exceedance of the 1-hour objective is unlikely where the annual mean concentration is below 60 µg/m^3.

Table 6.10: Predicted Annual Mean NO_2 Concentrations at the Site (µg/m^3)

Receptor Number	2026 With Proposed Development	Impact Significance
A	15.9	Negligible
B	14.6	Negligible

Receptor Number	2026 With Proposed Development	Impact Significance
C	15.8	Negligible
D	14.9	Negligible

Table 6.11: Predicted Annual Mean NO_2 Concentrations at Existing Receptors ($\mu g/m^3$)

Receptor Number	2026 base	2026 With Proposed Development	Increase due to Proposed Development	Impact Significance
1	17.0	16.9	-0.1	Negligible
2	18.8	18.7	-0.1	Negligible
3	20.4	20.3	-0.1	Negligible
4	15.7	15.7	0.0	Negligible
5	16.4	16.7	0.3	Negligible
6	15.7	16.0	0.3	Negligible
7	17.3	17.4	0.1	Negligible
8	17.3	17.4	0.1	Negligible
9	19.8	19.9	0.1	Negligible
10	19.7	19.8	0.1	Negligible
11	25.5	25.7	0.2	Negligible
12	21.9	24.3	2.4	Negligible
13	15.6	16.2	0.6	Negligible
14	14.0	14.6	0.6	Negligible
15	14.0	14.0	0.0	Negligible
16	16.1	15.8	-0.3	Negligible
17	15.2	15.0	-0.2	Negligible
18	15.7	15.4	-0.3	Negligible
19	16.6	16.4	-0.2	Negligible
20	14.0	14.0	0.0	Negligible
21	18.6	19.4	0.8	Negligible
22	15.9	15.8	-0.1	Negligible
23	17.8	17.9	0.1	Negligible
24	16.3	16.8	0.5	Negligible
25	18.1	18.4	0.3	Negligible
26	17.4	17.5	0.1	Negligible
27	16.0	16.1	0.1	Negligible

Receptor Number	2026 base	2026 With Proposed Development	Increase due to Proposed Development	Impact Significance
28	17.5	17.8	0.3	Negligible
29	18.5	18.4	-0.1	Negligible
30	16.4	16.8	0.4	Negligible
31	20.2	20.4	0.2	Negligible

PM_{10} Concentrations

6.5.28 Predicted annual mean PM_{10} concentrations at the Site are presented below in Table 6.12 and concentrations predicted at the existing receptors are provided in Table 6.13.

6.5.29 The predicted annual mean PM_{10} concentrations are 'well below' the 40 µg/m^3 objective across the Site; therefore, the impact on future occupants of the Site with regard to air quality will be negligible.

6.5.30 Annual mean PM_{10} concentrations at the existing receptors are also predicted to fall 'well below' the annual mean objective under both future scenarios.

6.5.31 The maximum contribution from the Proposed Development is predicted to be 0.6 µg/m^3, again at receptor 12, which is considered to be a small change in concentrations. As annual mean concentrations remain 'well below' the objective, with the Proposed Development in place, a small change in concentrations is considered to be of negligible significance.

6.5.32 The number of exceedances of 50 µg/m^3, as a 24-hour mean PM_{10} concentration, has been calculated from the annual mean following the approach set out by DEFRA in LAQM.TG(09):

A = -18.5 + 0.00145 x annual mean3 + (206/annual mean)

where A is the number of exceedances of 50 µg/m^3 as a 24-hour mean PM_{10} concentration.

6.5.33 Based on the above approach, the maximum number of days >50 µg/m^3 PM_{10} is between 0-8, with a maximum increase in the number of days exceeding the objective of 1 as a result of the Proposed Development. The objective for this pollutant permits up to 35-days per annum and therefore exceedence of this

objective is highly unlikely. The impact of the Proposed Development on the maximum number of exceedances of the 24-hour mean PM_{10} objective is negligible.

Table 6.12: Predicted Annual Mean PM_{10} Concentrations at the development site (µg/m^3)

Receptor Number	2026 With Proposed Development	Impact Significance
A	16.9	Negligible
B	16.5	Negligible
C	17.5	Negligible
D	16.8	Negligible

Table 6.13: Predicted Annual Mean PM_{10} Concentrations at Existing Receptors ($\mu g/m^3$)

Receptor Number	2026 base	2026 With Proposed Development	Increase due to Proposed Development	Impact Significance
1	17.9	17.8	-0.1	Negligible
2	18.8	18.7	-0.1	Negligible
3	19.5	19.5	0.0	Negligible
4	17.3	17.3	0.0	Negligible
5	17.8	17.8	0.0	Negligible
6	17.4	17.4	0.0	Negligible
7	18.3	18.3	0.0	Negligible
8	18.3	18.3	0.0	Negligible
9	19.7	19.7	0.0	Negligible
10	19.6	19.7	0.1	Negligible
11	23.0	23.1	0.1	Negligible
12	20.9	21.5	0.6	Negligible
13	17.4	17.5	0.1	Negligible
14	16.5	16.8	0.3	Negligible
15	16.5	16.5	0.0	Negligible
16	17.5	17.3	-0.2	Negligible
17	17.1	16.9	-0.2	Negligible
18	17.3	17.1	-0.2	Negligible
19	17.7	17.6	-0.1	Negligible
20	16.5	16.5	0.0	Negligible
21	19.0	19.1	0.1	Negligible
22	17.4	17.4	0.0	Negligible
23	18.3	18.3	0.0	Negligible
24	17.7	17.8	0.1	Negligible
25	18.5	18.6	0.1	Negligible
26	18.1	18.1	0.0	Negligible
27	17.4	17.5	0.1	Negligible
28	18.4	18.3	-0.1	Negligible
29	18.7	18.5	-0.2	Negligible
30	17.7	17.8	0.1	Negligible
31	19.4	19.5	0.1	Negligible

$PM_{2.5}$ Concentrations

6.5.34 Predicted annual mean $PM_{2.5}$ concentrations at the Site are presented below in Table 6.14, with concentrations predicted at the existing receptors given in Table 6.15.

6.5.35 The predicted annual mean $PM_{2.5}$ concentrations are 'well below' the 25 µg/m^3 objective across the Site. The impact on future occupants is therefore considered to be negligible.

6.5.36 Annual mean concentrations at the existing receptors are also predicted to be 'well below' the objective limit.

6.5.37 The maximum contribution from the Proposed Development is predicted to be 0.3 µg/m^3, again at receptor 12, which is a small change in concentrations (Table 6.4) with a negligible significance.

Table 6.14: Predicted Annual Mean $PM_{2.5}$ Concentrations at the Site (µg/m^3)

Receptor Number	2026 With Proposed Development	Impact Significance
A	11.5	Negligible
B	11.1	Negligible
C	11.4	Negligible
D	11.2	Negligible

Table 6.15: Predicted Annual Mean $PM_{2.5}$ Concentrations at Existing Receptors (µg/m^3)

Receptor Number	2026 base	2026 With Proposed Development	Increase due to Proposed Development	Impact Significance
1	11.8	11.7	-0.1	Negligible
2	12.2	12.2	0.0	Negligible
3	12.7	12.6	-0.1	Negligible
4	11.4	11.4	0.0	Negligible
5	11.7	11.7	0.0	Negligible
6	11.5	11.5	0.0	Negligible
7	12.0	12.0	0.0	Negligible
8	12.0	12.0	0.0	Negligible

Receptor Number	2026 base	2026 With Proposed Development	Increase due to Proposed Development	Impact Significance
9	12.7	12.8	0.1	Negligible
10	12.7	12.7	0.0	Negligible
11	14.5	14.6	0.1	Negligible
12	13.4	13.7	0.3	Negligible
13	11.5	11.5	0.0	Negligible
14	11.0	11.1	0.1	Negligible
15	11.0	11.0	0.0	Negligible
16	11.5	11.4	-0.1	Negligible
17	11.3	11.2	-0.1	Negligible
18	11.4	11.3	-0.1	Negligible
19	11.7	11.6	-0.1	Negligible
20	11.0	11.0	0.0	Negligible
21	12.3	12.4	0.1	Negligible
22	11.5	11.5	0.0	Negligible
23	12.0	12.0	0.0	Negligible
24	11.7	11.7	0.0	Negligible
25	12.1	12.1	0.0	Negligible
26	11.8	11.9	0.1	Negligible
27	11.5	11.5	0.0	Negligible
28	12.0	12.0	0.0	Negligible
29	12.2	12.1	-0.1	Negligible
30	11.7	11.7	0.0	Negligible
31	12.6	12.7	0.1	Negligible

6.6 MITIGATION MEASURES

Construction stage

6.6.1 It is proposed that the following 'best practice' measures be implemented as appropriate during the construction stage:

- ensure effective site planning, locating machinery and dust-causing activities away from sensitive receptors;
- erect solid screens or barriers around the site boundary, where possible;
- vehicles carrying loose aggregate and workings should be sheeted at all times;
- all vehicles should switch off engines when not in use i.e. no idling vehicles should occur at the Site;
- stockpiles should be kept for the shortest time possible and, if necessary, the use of sprinklers and hoses for dampening of exposed soil and materials should be employed;
- observation of wind speed and direction prior to conducting dust-generating activities to determine the potential for dust nuisance to occur, avoiding potentially dust-generating activities during periods when the wind direction may carry dust into sensitive areas and avoiding dust-generating operations during periods of high or gusty winds;
- stockpiles of soils and materials should be located as far as possible from sensitive properties, taking account of prevailing wind directions and seasonal variations in the prevailing wind;
- completed earthworks should be covered or vegetated as soon as is practicable;
- regular inspection of local highways and site boundaries to check for dust deposits and, if necessary, removal and cleaning of any such deposits;
- visual inspection of the site perimeter to check for dust deposition (evident as soiling and marking) on vegetation, cars and other objects and taking remedial measures if necessary;
- minimise surface areas of stockpiles (subject to health and safety and visual constraints regarding slope gradients and visual intrusion) to reduce the area of surfaces exposed to wind pick-up;
- ensure concrete batcher, where used, has a permit to operate and is operated in accordance with Process Guidance Note 3/1 (04);

- use of dust-suppressed tools for all operations;
- ensuring that all construction plant and equipment is maintained in good working order; and
- no unauthorised burning of any material anywhere on site.

6.6.2 Construction vehicles should be kept clean and sheeted when on public highways. Timing of large-scale vehicle movements to avoid peak hours on the local road network will also be beneficial.

6.6.3 It is recommended that liaison with the local authority be maintained throughout the construction process, and any incidents which lead to excessive elevation of dust deposition and/or PM_{10} concentrations at neighbouring sensitive receptors are reported to the Environmental Health Department. If complaints are received from local residents, these will be documented in a diary or log held on site by the Site Manager. A nominated member of the construction team (e.g. the Site Manager) will also act as a point of contact for residents who may be concerned about elevated deposition of dust.

6.6.4 The overall sensitivity of the Site is considered to be medium, based on the IAQM guidance. The overall significance of potential dust impacts based on this guidance and professional judgement is considered to be negligible, following the implementation of appropriate and best practice mitigation measures as detailed above.

Post-completion stage

6.6.5 Annual mean NO_2, PM_{10} and $PM_{2.5}$ concentrations are predicted to be 'well below' the objective of 40 $\mu g/m^3$ at all locations. The Proposed Development is predicted to result in a negligible change in pollution concentrations; therefore no mitigation measures are considered necessary.

6.7 RESIDUAL EFFECTS

Construction stage

6.7.1 The greatest potential for dust nuisance problems to occur will generally be within 200m of the construction site perimeter. There may be limited incidences of increased dust deposited on property beyond this distance.

6.7.2 By following the mitigation measures outlined within this chapter, the impact will be substantially minimised. Residual impacts are considered to be negligible.

Post-completion stage

6.7.3 The Proposed Development is predicted to cause a negligible impact on local air quality; therefore the residual impacts will be negligible.

Summary of effects

5.7.3 The effects identified are summarised in Table 6.16 below:

Table 6.16: Summary of effects

Potential effect	Significance (pre-mitigation)	Mitigation measure	Significance of residual effect
Construction stage			
Dust/PM_{10} Impacts during demolition	**Low Risk**	Best practice measures as set out in current IAQM guidance	**Negligible**
Dust/PM_{10} Impacts during earthworks	**Medium/High Risk**	Best practice measures as set out in current IAQM guidance	**Negligible**
Dust/PM_{10} Impacts during construction	**High Risk**	Best practice measures as set out in current IAQM guidance	**Negligible**
Dust/PM_{10} Impacts due to trackout	**Low Risk**	Best practice measures as set out in current IAQM guidance	**Negligible**
Post-completion stage			
Impact on local air quality (NO_2, PM_{10}	**Negligible**	No mitigation required	**Negligible**

and $PM_{2.5}$)			
Exposure of future occupants to poor air quality	**Negligible**	No mitigation required	**Negligible**

6.8 CUMULATIVE EFFECTS

6 The cumulative air quality effects of the development have been taken into account through the inclusion of the allocations within the transport modelling work. This is fully identified in the TA attached as **appendix 5.1** to this ES.